

Solution 3.2

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$ for any non-zero vector $\vec{v}$.
- $\vec{i} \cdot \vec{j} = 0$ since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$.